

Arunabha Majumder

Mechanical Engineer | Lab Hardware, Rapid Prototyping & Systems Integration (SolidWorks)

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SUMMARY

Mechanical engineer who designs and builds lab hardware: SolidWorks concepts turned into testable prototypes in days through 3D printing and machining, then measured, adjusted and rebuilt. Integrates actuators, sensors and electronics into single working systems, with a strong sense for geometry, stiffness and how parts fit together. Curious about the physics behind the instrument.

KEY RESULTS

- **Days, not months:** idea to testable hardware, built and iterated by hand
- **<10% RMS error** on a geometry-based force measurement, validated against a force sensor
- **>25% energy reduction** from a new mechanical concept (Best Paper Award, IFToMM ISRM 2026)

EXPERIENCE

Research Engineer – Mechanical Design & Rapid Prototyping of Lab Hardware (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Turned new ideas into testable hardware quickly by choosing the right process for each part (3D printing on in-house printers, machined parts, off-the-shelf components), then building, testing and iterating the designs myself.
- Integrated the full experimental setup into one working system (actuators, motor, encoder, force sensors, DAQ and embedded control) by designing the mechanical assemblies in SolidWorks so every component fitted together, physically and functionally.
- Kept load-carrying structures stiff and light, checked with FEA for stiffness, strength and weight, by choosing materials and geometries for each part before manufacture.
- Achieved below 10% RMS error in force measurement on a 5-bar planar parallel robot, validated against a force sensor, by modelling its geometry and kinematics (Jacobian) and variable-stiffness actuators; applied the method in lab measurements of human arm impedance.
- Cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026) by designing a new hybrid actuation concept for a shoulder exoskeleton: a parallel spring working with a 6 Nm motor.
- Documented designs so others could build on them (3D models, drawings, part specifications), working closely with engineers and scientists across disciplines and mentoring Bachelor's students (2023 – 2025).

MTech Researcher – Pneumatic Actuator Test Setups (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Built a full test setup from scratch by fabricating pneumatic artificial muscles from raw materials and integrating valves, pressure sensors and DAQ.
- Reduced position-tracking error to 0.3–0.78% with a neural-network-based controller, outperforming classical PID.

THINGS I BUILT

- **Exoskeleton prototype (PhD):** Hybrid-actuated joint, test rigs and full sensor/electronics integration.
- **5-bar parallel robot setup (PhD):** Force measurement from actuator models, used in lab experiments.
- **Pneumatic test bench (MTech):** Actuators and pressure test setup built from raw materials.

- **Go-Kart (2017):** Designed, fabricated and assembled from the ground up.

SKILLS

- **CAD:** SolidWorks (CSWA), Fusion 360, assemblies, drawings, part specifications
- **Rapid prototyping:** 3D printing (Bambu Lab, Ultimaker Cura), machined parts, off-the-shelf integration, hands-on assembly
- **Analysis:** FEA (stiffness, strength, weight), kinematics and geometry, material and geometry selection
- **Integration & software:** Sensors, encoders, motors, DAQ, embedded control, MATLAB/Simulink, Python
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Two peer-reviewed publications: ISRM 2026 (Springer) and IEEE CONECCT 2022

REFERENCES

Available upon request.